

Geometry of Splats

Definition (Bures-Wasserstein Manifold)

Fix an isotropic, centered density $\rho(\cdot) \in \mathcal{P}_2(\mathbb{R}^d)$ and let $\text{BW}_\rho(\mathbb{R}^d)$ contain all *affine pushforwards*,

$$\text{BW}_\rho(\mathbb{R}^d) = \{(A(\cdot) + b)_\# \rho : b \in \mathbb{R}^d, A \in \mathbb{R}^{d \times d}\}$$

endowed with the Wasserstein metric. This manifold is *geodesically convex* and *invariant to choice of ρ* .

“Splats” are elements of $\text{BW}_\rho(\mathbb{R}^d)$ and ρ is the “mother splat.”

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Definition (Splat Measures, Splat Functions)

We call $\mathcal{S}_{p,d} := \mathcal{P}(\mathbb{R}^p \times \text{BW}_\rho(\mathbb{R}^d))$ the family of *splat measures*. A function f is a *splat function* if there exists $\mu \in \mathcal{S}_{p,d}$ so that:

$$f(x) = f_\mu(x) := \mathbb{E}_{v, \bar{\rho} \sim \mu}[v \bar{\rho}(x)]$$